HDD → Alternative → SSD. → Flash NAND

↓ 2 type

internal (fixed with system) PCIe

External (portable) USB (portable)

External → portable.

Solid → nand / nor ckt.

SSD

Tech: NAND ckt

Allocate: Fast (i/o operation)

cost: ↑

capacity: 1TB

SSD – 8K
HDD – 4K

Size → small (chip)

Robustness → More Robust.

Power Consumption: SSD better

Heat production: SSD cooler.

↓ external mem.

HDD.

magnetic

Slow (inversing operation Head / Arm)

↓

large (Box)

mismatch.

can't be accessed directly.

CPU

OS

↳ File explorer. (just fetch the file name & properties of SSD

↓

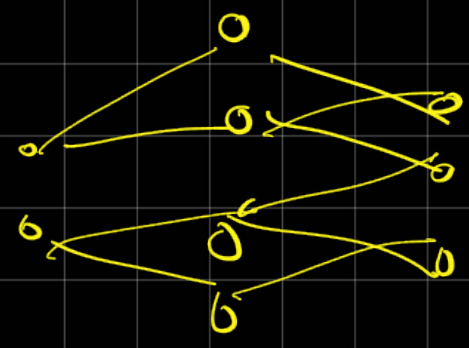
I/O drivers.

I/O module.

I/O device.

data scientist
Analyst.

bios → permanently store instr.



Controller,

Maintain communication with host system

|| underlying properties

execute some instr.

↓

stored in

(Hardwired programming)

↓

Firmware.

CD speed faster

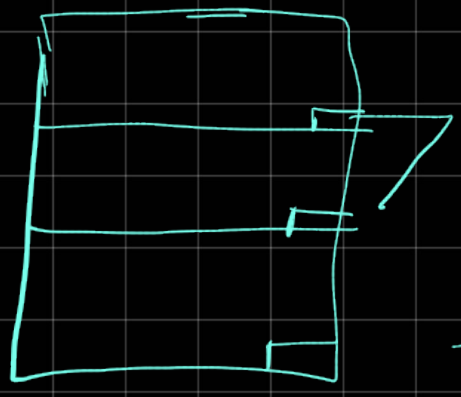
Write operation: more time.

can't directly update anything.

previous content erased totally

the new written

Issues → slow (faster than HDD)



small portion vacant

so full data can't be stored

together. So strip & store

here & there.

Soln: ① Over-provision

when write any component →

precopy

② Trim:

when SSD not working

Drivers will defragmentation.



(2) write count. (max) → 1,00,000
↓
Computers damaged.

Optical property:

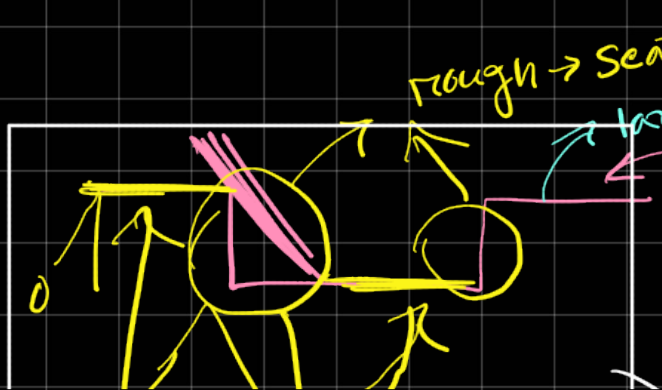
CD

Compact Disk ↔ light/Ray used → laser.



↳ Shiny surface ce using light.
Reflection intensity.

smooth surface → high reflection
rough " → low "

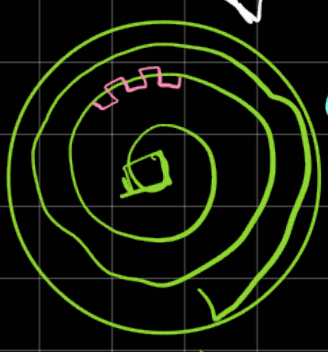




coated with Al or Gold for shiny surface

HDD & CD both round

how to store → track



↳ single spiral track

Sector → block

CD ROM → RAM/ROM variation?

CD-R → no erasing. → Permanent

CD-RW → not Permanent.

WORM

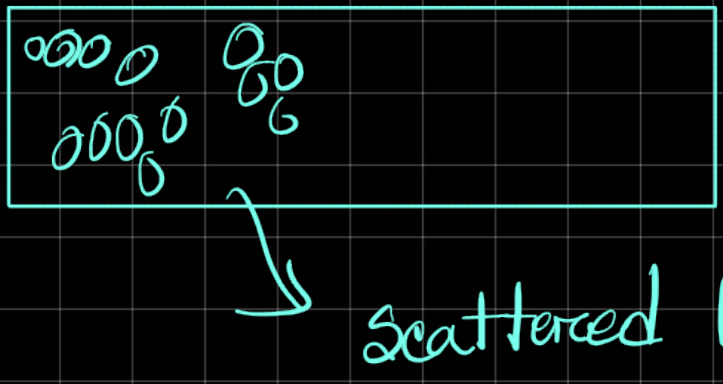
write once Read Many.

CD-R → CD rewritable.

special features will produces. when lands & pits are all impenetrated? then done.

change property of layer. Dye layer

maintain/control → crystaline
→ Amorphous
> molecular level modification.



Scattered light.